Math 101, HW 5 name:

Inverse Functions!

(3 points) Given $f(x)$ below, calculate the equation for the inverse function, $f^{-1}(x)$

Show all steps for full credit!

$$f(x) = 2x - 6$$

$f^{-1}(x) =$

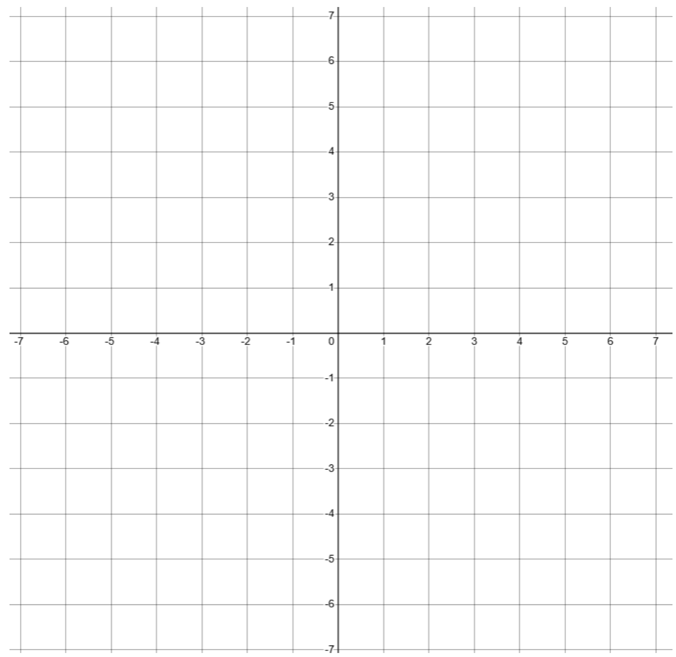
(2 points) Short Answer

How are the **equations** of $f(x)$ and $f^{-1}(x)$ related?

How are the **graphs** of $f(x)$ and $f^{-1}(x)$ related?

(5 points) Graph the original function $f(x)$ and the inverse function $f^{-1}(x)$ on the same grid below.

Be sure to include at least 3 reference points and label your graphs for full points 🙂



Exponential Functions!

You fell in a Mad River with flesh eating bacteria. Unfortunately, 950 bacterias got in through a open wound in your leg. The flesh eating bacteria population grows at a continuous rate of 8% per minute.

a) (4 points) Write the function equation for the number of bacteria you harbor. Use the template $A(t) = Pe^{rt}$

starting value: $P =$_____

rate of increase/decrease: $r =$ ______

$A(t) = Pe^{rt} =$

b) 6 points)You need to drive back to town and it will take 2 hours. How many bacterias will you have after 2 hours? *Note: Two hours = 120 minutes!*

(notes) 3.7 Inverse Functions

Original function: $f(x)$
Inverse function: $f^{-1}(x)$

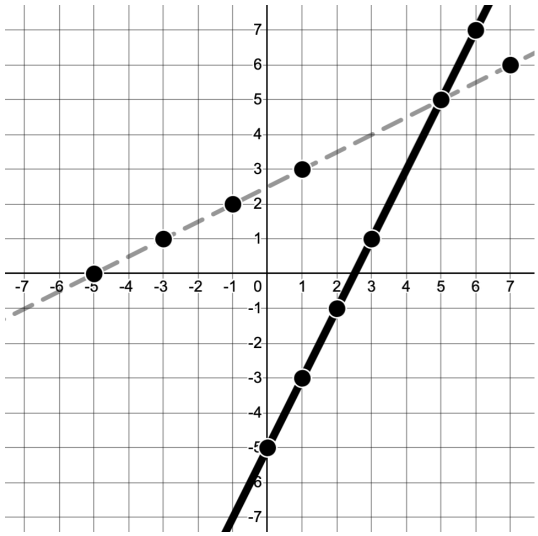
Key Takeaways

1. Algebra: when you apply the inverse operation, you get back the identity, x
2. Graphically: when you plot a function $f(x)$ and its inverse function $f^{-1}(x)$, the two functions are graphically symmetric over the identify function, $y = x$!
3. Functions need to be one-to-one to have an inverse. If not, we need to restrict the domain of $f(x)$.

Algebraic check!

$f(x)$	$f^{-1}(x)$	$f^{-1}(f(x)) = x$? (check if we get the identify, x)
$x + 2$		
$2x$		
x^2		
2^x	$log_2(x)$	$f^{-1}(f(x)) = log_2(f(x)) = log_2(2^x) = x$✓
$2x - 5$	$\frac{x+5}{2}$	$f^{-1}(f(x)) = \frac{f(x) + 5}{2}$

Graph check!



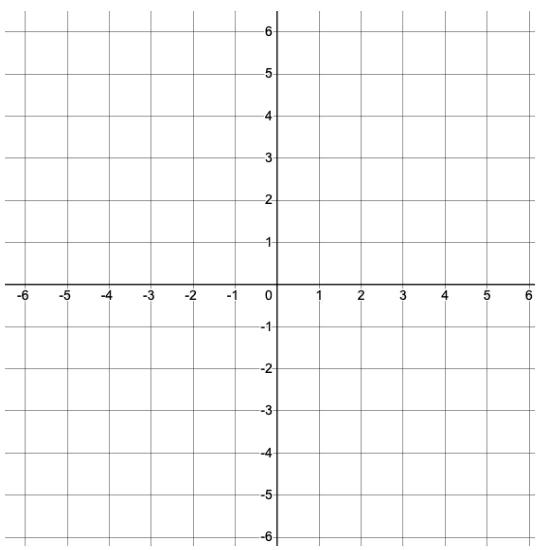
Solid: $f(x) = 2x - 5$

Dashed: $f^{-1}(x) = \frac{x+5}{2}$.

original function	inverse function
$(x, f(x))$	$(x, f^{-1}(x))$
$(0, -5)$	$(-5, 0)$
$(1, -3)$	$(-3, 1)$
$(2, -1)$	$(-1, 2)$
$(3, 1)$	$(1, 3)$
$(5, 5)$	$(5, 5)$
$(6, 7)$	$(7, 6)$

First, graph $f(x)$ and $f^{-1}(x)$.

Then, fill in the table with the coordinate values for each of the functions.

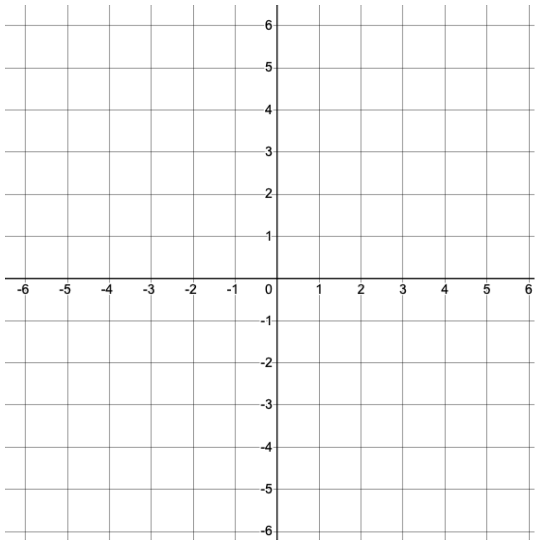


Graph $f(x) = 2x$

Graph $f^{-1}(x) = \frac{1}{2}x$

original function	inverse function
$(x, f(x))$	$(x, f^{-1}(x))$
$(-2, \qquad)$	$(\qquad , -2)$
$(-1, \qquad)$	$(\qquad , -1)$
$(0, \qquad)$	$(\qquad , 0)$
$(1, \qquad)$	$(\qquad , 1)$
$(2, \qquad)$	$(\qquad , 2)$

Graph check!



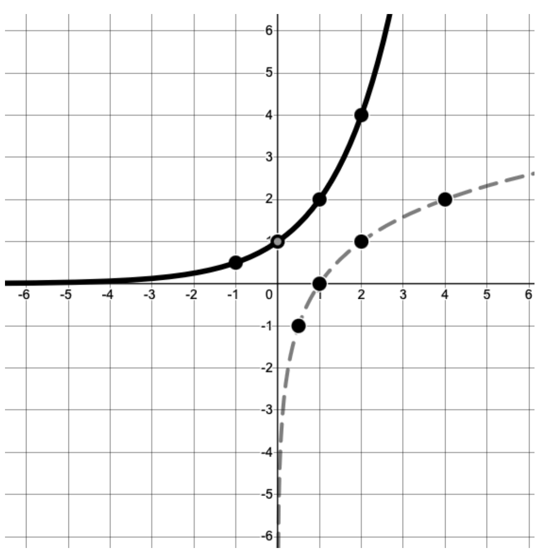
What is odd about this one???

Graph $f(x) = x^2$

Graph $f^{-1}(x) = \sqrt{x}$.

original function	inverse function
$(x, f(x))$	$(x, f^{-1}(x))$
$(-2, \quad\quad)$	$(\quad\quad , -2)$
$(-1, \quad\quad)$	$(\quad\quad , -1)$
$(0, \quad\quad)$	$(\quad\quad , 0)$
$(1, \quad\quad)$	$(\quad\quad , 1)$
$(2, \quad\quad)$	$(\quad\quad , 2)$

Use the graph to fill in the tables with indicated points



Solid: $f(x) = 2^x$

Dashed: $f^{-1}(x) = log_2(x)$.

original function	inverse function
$(x, f(x))$	$(x, f^{-1}(x))$
$(\quad , \quad)$	$(\quad , \quad)$
$(\quad , \quad)$	$(\quad , \quad)$
$(\quad , \quad)$	$(\quad , \quad)$
$(\quad , \quad)$	$(\quad , \quad)$
$(\quad , \quad)$	$(\quad , \quad)$

Discuss: how are the graphs of $f(x)$ and $f^{-1}(x)$ related? How are the coordinate points related?

Practice with tables

Question 2

Use the table below to fill in the missing values.

x	$f(x)$
0	7
1	1
2	9
3	4
4	8
5	6
6	3
7	0
8	5
9	2

answers: a)4, b) 5, c)3, d) 6

a. $f(3) =$

b. If $f(x) = 3$, then $x =$

c. $f^{-1}(4) =$

d. If $f^{-1}(x) = 5$, then $x =$

Solving for an inverse

To solve for the inverse, swap $y \leftrightarrow x$. Then solve for y.

Example. Solve for $f^{-1}(x)$

$$\begin{aligned} f(x) &= 7x + 9 \\ y &= 7x + 9 \\ &\text{inverse: } y \leftrightarrow x \\ x &= 7y + 9 && \to \text{now solve for } y \\ x - 9 &= 7y + 9 - 9 && \to \text{subtract } 9 \\ x - 9 &= 7y \\ \frac{x-9}{7} &= \frac{7y}{7} && \to \text{divide both sides by } 7 \\ \frac{x-9}{7} &= y && \implies f^{-1}(x) = \frac{x-9}{7} \end{aligned}$$

Practice, solve for $f^{-1}(x)$

1. $f(x) = 14 - x$
2. $f(x) = 2x + 5$
3. $f(x) = (x - 6)^2$
   (......is this function one-to-one and if not, how can we **restrict the domain** to just the non-decreasing section?)
4. $f(x) = \frac{1}{x+10}$
5. $f(x) = 12 + \sqrt[3]{x}$